

Simple critical zeros in short intervals: stability, pressure, and parity

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CAOS Research program

github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Let $c(\theta) = 2 - \theta/2 - \cot(\theta/\sqrt{2})/\sqrt{2}$, and let θ_0 be its unique zero in $(0, 1)$. We combine the finite stable rank–trace inequality underlying recent zeta-zero results with a classical positive density of distinct odd-multiplicity critical zeros. If $\kappa > 0$ is one fixed Selberg density constant packed into every longer interval, then, for each fixed $0.51 < \theta < 1$, the lower asymptotic simple-critical proportion in $(T, T + T^\theta]$ is at least $\max\{0, c(\theta), (c(\theta) + 2\kappa)/3\}$. Consequently it is positive at some fixed exponent strictly below θ_0 . A separate multiplicity identity gives the correct distinct-zero companion, including a proportion greater than one half at another exponent below θ_0 . The new exponents are expressed in terms of the unevaluated classical constant; no new decimal exponent is claimed. We also strengthen the entire positive cosine curve by an odd-frame pressure transfer. A two-variable interval certificate at $\theta = 3/4$ yields the simple-critical bound $0.4190878881701117\dots$ and distinct bound $0.7095439440850558\dots$. The finite stability inequality, the classical sign-change theorems, and the global pressure methods are prior work. Wang’s fixed-test pair-correlation theorem supplies the arithmetic input; all smoothing and support limits follow the height limit. The qualitative parity result requires no numerical zero sampling or kernel optimization. No global record, effective starting height, or solution of the Riemann hypothesis is claimed.

Summary of results

The central result is a parity transfer from classical odd-multiplicity critical zeros to simple critical zeros. Retaining multiplicity excess in the known finite stability inequality yields a qualitative extension below the zero of Wang’s cosine bound. This changes the admissible interval-length range rather than only a numerical proportion. The explicit dependence on the classical density constant and the separate distinct-zero argument are proved below.

Further results are the direct finite Hilbert transfer with its explicit compact energy bound, the odd-frame pressure theorem, its stronger consequence throughout the positive short-interval

curve, an odd-frame bound from the compact finite certificate, and a certified two-variable pressure bound. The following table compares the separate quantitative outcomes; the parity range theorem has no certified decimal exponent and is not represented by a numerical row.

Verification. For the pressure bounds and the parity deduction (computations EXP-003 and EXP-004 [15, 16]), the verification consists of exact finite checks, complete 256-bit replays of the interval certificates, checks of the imported statements against their primary sources, and independent step-by-step checks of the proofs, including an independent recomputation of every census row. It is not external peer review or end-to-end formal verification.

Bound at $\theta = 3/4$	Simple-critical proportion	Gain above Wang
Wang	0.4190750129754...	0
Direct transfer, Cor. 1.5	0.4190768284253...	$1.8154498796... \times 10^{-6}$
Odd-frame bound, Cor. 1.6	0.4190777360205...	$2.7230451445... \times 10^{-6}$
Two-variable pressure certificate, Cor. 1.6	0.4190878881701...	$1.2875194687... \times 10^{-5}$

1 Statement and relation to prior work

Let $N(T, H)$ count the nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $T < \gamma \leq T + H$, including multiplicity. Let $N_0^s(T, H)$ count the simple zeros among them with $\beta = 1/2$, and let $N^d(T, H)$ count distinct complex zeros, each once. Let $N_{\text{odd}}(T, H)$ count distinct critical zeros of odd multiplicity, also each once. It does not count their multiplicity. Every proportion below has denominator $N(T, H)$.

The recent argument discovered by Claude and communicated by Alpöge and Furman [1] made the optimized Montgomery–Taylor simple-critical proportion $0.672500703679...$ unconditional. Lamzouri [2] replaced its finite-compression machinery by a Hilbert-space inequality for conjugation-invariant multisets. Wang [3] established the needed short-interval pair-correlation theorem and deduced, for fixed $0 < \theta < 1$,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c(\theta), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c(\theta)}{2}, \quad (1)$$

where

$$c(\theta) = 2 - \frac{\theta}{2} - \frac{1}{\sqrt{2}} \cot(\theta/\sqrt{2}). \quad (2)$$

The derivative is $c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) > 0$. Since $c(\theta) \rightarrow -\infty$ as $\theta \downarrow 0$ and $c(1) > 0$, there is a unique zero $\theta_0 \in (0, 1)$, numerically $\theta_0 \approx 0.550193964744154...$, as printed by Wang. The proof below uses its defining equation and rational comparisons, not a numerical root certificate.

The stable rank–trace inequality, convex Gram defect, pinching method, and cosine-root non-additivity used here are prior work in the `ainta` repository [8]. Global pressure windows, larger frames, nonuniform pair capacities, spectral envelopes, and combinations of certificates are developed in [9, 11, 12]. The alternating schedule used below is a simple choice within this established framework. We do not claim these ingredients or an improved global record. Our contributions are their direct finite-Hilbert transfer, an explicit quantitative three-point bound, a strict refinement of Wang’s positive short-interval curve, and two consequences. First, classical odd-multiplicity density and the retained multiplicity excess extend the positivity range qualitatively. Second, odd-frame pressure strengthens the positive curve quantitatively, with independently replayed numerical inputs. The finite proof keeps off-line signs, multiplicities, and singular Gram matrices explicit.

Selberg's positive density theorem at lengths $T^{1/2+\epsilon}$ is reproduced as Theorem B in Karatsuba's primary article [4]. Karatsuba's own theorem works at $T^{27/82+\epsilon}$. These count distinct odd-order critical zeros. They do not already assert a positive simple-zero proportion. Steuding's later simple-critical theorem gives the explicit exponent 0.552 [5, 6]; Wang gives the smaller explicit threshold θ_0 . The combination below uses only the convenient fixed classical seed $\alpha = 51/100$, and neither improves Karatsuba's odd-zero theorem nor computes a replacement decimal for θ_0 .

The labels [D] and [MV] distinguish mathematical deductions from finite machine-verifiable statements. The following main range result is a deduction from the stated primary analytic inputs. Exact finite checks verify its algebra; they do not replace those inputs or prove all-height statements by enumeration.

Theorem 1.1 (Parity transfer and qualitative range extension, D). *Put $\alpha = 51/100$. There exists a constant $\kappa > 0$ such that, for every fixed $\alpha < \theta < 1$,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2\kappa}{3} \right\}, \quad (3)$$

$$\liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ \kappa, \frac{1 + c(\theta)}{2}, \frac{3 + 2c(\theta) + \kappa}{6} \right\}. \quad (4)$$

One fixed κ works for all these fixed exponents; the eventual height may depend on θ . Define

$$\delta = \min \left\{ \frac{\theta_0 - \alpha}{2}, \frac{\kappa}{4} \right\} > 0, \quad \theta_1 = \theta_0 - \delta. \quad (5)$$

Then $\alpha < \theta_1 < \theta_0$ and the lower simple-critical proportion at every fixed $\theta_1 \leq \theta < 1$ is at least $\kappa/3 > 0$. Separately, set

$$\delta_{\text{half}} = \min \left\{ \frac{\theta_0 - \alpha}{2}, \frac{\kappa}{16} \right\}, \quad \theta_{\text{half}} = \theta_0 - \delta_{\text{half}}. \quad (6)$$

For every fixed $\theta_{\text{half}} \leq \theta < 1$, the lower distinct-zero proportion is at least $1/2 + \kappa/12$.

The distinct claim is greater than one half on some shorter intervals; it is not a new exponent for merely positive distinct density, which already follows from the classical odd-zero theorem. Theorem 1.1 does not give an effective value of κ , a numerical value of either new exponent, or a uniform theorem when θ varies with T .

Theorem 1.2 (Explicit three-point refinement, D). *Fix $\theta_0 < \theta < 1$ and $R > 2/c(\theta)$. Define*

$$\begin{aligned} a &= \theta/\sqrt{2}, & q &= a \tan a, & M &= \pi\theta R, \\ B &= q(1 + M^2/a^2), \\ b &= \min \left\{ \frac{q}{2}, \frac{q^4}{4(q+M)^2(M+3q/2)} \right\}, & \delta &= \frac{1}{2} \min\{1, 2(b/B)^2\}. \end{aligned} \quad (7)$$

Then $0 < \delta \leq 1/2$, and, with

$$c_*(\theta, R) = c(\theta) + \frac{\delta\{c(\theta) - 2/R\}}{3 - \delta} > c(\theta), \quad (8)$$

we have

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c_*(\theta, R), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c_*(\theta, R)}{2}. \quad (9)$$

In particular, $R = 4/c(\theta)$ gives a fully specified positive improvement.

Theorem 1.3 (Odd-frame pressure refinement, D). *Fix $\theta_0 < \theta < 1$. Suppose $p, \varepsilon > 0$ and the cosine kernel K_θ of (30) satisfies*

$$2\{K_\theta(u)^2 + K_\theta(v)^2 + K_\theta(u+v)^2\} + p(u+v) \geq \varepsilon \quad (u, v \geq 0). \quad (10)$$

For every fixed integer $k \geq 1$ with $k\varepsilon \leq 1$, put

$$c_{\text{odd}} = c(\theta) + \frac{k\{\varepsilon c(\theta) - 2p\}}{2k + 1 - k\varepsilon}. \quad (11)$$

Then the lower asymptotic simple-critical and distinct proportions are at least c_{odd} and $(1+c_{\text{odd}})/2$, respectively. The gain is strictly positive when $\varepsilon c(\theta) > 2p$.

Corollary 1.4 (Stronger entire positive curve, D). *For the analytic δ in (7), one has $0 < \delta < 1/4$. Every integer $k > 1$ with $k\delta \leq 1$ strengthens Theorem 1.2 by replacing its gain coefficient $\delta/(3-\delta)$ with $k\delta/(2k+1-k\delta)$. In particular $k=2$ is available for every fixed $\theta_0 < \theta < 1$.*

The conservative gap in Theorem 1.2 requires no numerical search. A stronger finite certificate gives the following example.

Corollary 1.5 (Compact-certificate example, D+MV). *Set $\theta = 3/4$, $R = 21/4$, $d = 1/7000$, and*

$$C_* := \frac{3c(3/4) - 2d/R}{3 - d} = 0.419076828425303996736665787527 \dots$$

Then the simple-critical proportion in (9) is at least C_ , and the distinct-zero proportion is at least*

$$\frac{1 + C_*}{2} = 0.709538414212651998368332893763 \dots$$

The improvement in the simple-critical proportion is 0.000001815449879663002112176828 \dots

Corollary 1.6 (Odd-frame bound and two-variable pressure certificate, D+MV). *At $\theta = 3/4$, the compact energy certificate of Section 7, with $d = 1/7000$ and $R = 21/4$, gives, with $k = 7000$,*

$$C_A = c(3/4) + \frac{c(3/4) - 8/21}{14000} = 0.419077736020568836833221725071 \dots \quad (12)$$

The all-gap pressure certificate of Appendix B has

$$p = \frac{1}{12500}, \quad \varepsilon = \frac{443239}{10^9}, \quad k = 2256, \quad 2k + 1 = 4513. \quad (13)$$

It yields

$$\begin{aligned} C_B &= c(3/4) + \frac{2256\{(443239/10^9)c(3/4) - 2/12500\}}{4513 - 2256(443239/10^9)} \\ &= 0.419087888170111727959091183775 \dots \end{aligned} \quad (14)$$

The respective distinct companions are

$$(1 + C_A)/2 = 0.709538868010284418416610862535 \dots,$$

$$(1 + C_B)/2 = 0.709543944085055863979545591887 \dots$$

The conclusion is for each fixed exponent and all sufficiently large heights in the asymptotic sense. We do not provide an effective starting height or uniformity as an exponent approaches its endpoint. The pressure corollaries themselves concern $\theta > \theta_0$; the distinct parity argument in Theorem 1.1 supplies the qualitative extension below θ_0 . The common analytic dependency is Wang's inspected preprint theorem, whose underlying pair-correlation method is due to Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh [10, 13].

2 A stable inequality for finite multisets

Let Z be a finite conjugation-invariant multiset in $\mathbb{C}$. We sum over its distinct support with positive integer multiplicities $m_z = m_{\bar{z}}$, and put $N = \sum_z m_z$. Choose a real even compactly supported $\eta \in L^2(\mathbb{R})$ with $\int \eta^2 = 1$, and define

$$v_z(u) = \eta(u)e^{-2\pi izu}, \quad K(t) = \widehat{\eta^2}(t), \quad \widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{-2\pi itu} du.$$

Every v_z belongs to L^2 because Z is finite and the support is compact. Work in the finite-dimensional span $\mathcal{H}$ of these vectors. Our inner product is $\langle v, w \rangle = \int \bar{v}w$, linear in the second variable; write $|v\rangle\langle w|$ for $h \mapsto v\langle w, h \rangle$.

Define the reflected rank-one operator

$$A = \sum_z m_z |v_z\rangle\langle v_{\bar{z}}|. \quad (15)$$

It is selfadjoint by conjugation invariance. Direct multiplication gives

$$\mathrm{tr} A = N, \quad Q := \|A\|_{\mathrm{HS}}^2 = \mathrm{tr} A^2 = \sum_{z,w} m_z m_w K(z-w)^2. \quad (16)$$

Indeed, $\langle v_{\bar{z}}, v_z \rangle = \int \eta^2 = 1$, while the trace of the product of the z and w terms, before multiplicity factors, is $K(w-z)K(z-w) = K(z-w)^2$, since K is even. The ordinary square in (16) is essential. Individual terms for complex arguments need not be positive or even real; their complete sum is a nonnegative operator norm.

Let $x_1 < \dots < x_s$ be the simple real points of Z . Let r be the number of distinct multiple real points and k the number of distinct nonreal conjugate pairs. Put

$$P = \sum_{j=1}^s |v_{x_j}\rangle\langle v_{x_j}| = VV^*, \quad G = V^*V = (K(x_i - x_j))_{i,j=1}^s, \quad C = A - P.$$

Then G is positive semidefinite with diagonal one and $\mathrm{tr} P = s$. For each nonreal pair put $g = (v_z + v_{\bar{z}})/2$ and $h = (v_z - v_{\bar{z}})/(2i)$. Its contribution to C is $2m_z(|g\rangle\langle g| - |h\rangle\langle h|)$. The form of C is nonpositive on the orthogonal complement of the r multiple-real vectors and k vectors g . Thus

$$n_+(C) \leq r + k, \quad N \geq s + 2r + 2k. \quad (17)$$

No distance of a nonreal point from the real axis is assumed.

Lemma 2.1 (Known stable rank–trace inequality). *Suppose $P = VV^*$ on a finite-dimensional Hilbert space, the s columns of V have norm one, C is selfadjoint, and $n_+(C) \leq d$. Define*

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases}$$

Then

$$\|P + C\|_{\mathrm{HS}}^2 \geq 4\mathrm{tr}(P + C) - 3s - 4d + \mathrm{tr} \Psi(V^*V). \quad (18)$$

Proof. This is the unit-column case of the stability inequality in [8, Section 2]. We give a spectral proof that keeps all zero eigenvalues. Let $m = \dim \mathcal{H}$ and let $p_1 \leq \dots \leq p_m$ be the eigenvalues of

P . If $d < m$, the min-max principle gives $\lambda_i(P + C) \leq p_{i+d}$ for $1 \leq i \leq m - d$: intersect the span of the first $i + d$ eigenvectors of P with a codimension-at-most- d subspace on which $C \leq 0$.

It follows that

$$\operatorname{tr}(P + C - 2I)^2 \geq \sum_{j=d+1}^m (2 - p_j)_+^2 \geq \sum_{j=1}^m (2 - p_j)_+^2 - 4d.$$

Here $0 \leq (2 - p_j)_+^2 \leq 4$. If $d \geq m$, the final right-hand side is nonpositive and the same inequality is immediate. For every $p \geq 0$, $(2 - p)_+^2 = \Psi(p) - 2p + 3$. The nonzero spectra of VV^* and V^*V agree, and $\Psi(0) = 1$, so

$$\operatorname{tr} \Psi(P) = \operatorname{tr} \Psi(V^*V) + m - s.$$

Substitute these identities and $\operatorname{tr} P = s$, and expand the left-hand square. All dimensional terms cancel and give (18). $\square$

Proposition 2.2 (Finite Hilbert stability transfer). *For the multiset and operator above, write $D(G) = \operatorname{tr} \Psi(G)$ and $D_Z = s + r + 2k$ for the number of distinct support points. Then*

$$s \geq 2N - Q + D(G), \quad D_Z \geq \frac{3N - Q + D(G)}{2}. \quad (19)$$

Proof. By Lemma 2.1 and (17), $Q \geq 4N - 3s - 4(r + k) + D(G)$. The multiplicity bound in (17) makes this at least $2N - s + D(G)$. For the second claim, the difference between $4N - 3s - 4(r + k)$ and $3N - 2D_Z$ is $N - s - 2r \geq 0$. $\square$

This proof retains the signed off-line remainder throughout. The positive Gram defect belongs only to the actual simple real points. The second inequality in (19), rather than elementary counting from a simple-zero proportion, will yield the distinct-zero companion.

2.1 Retaining multiplicity excess and parity

Keep the notation of Proposition 2.2; in this subsection k counts nonreal pairs, not frame length. Let O count the distinct real support points of odd multiplicity, and define the nonnegative excess

$$E = N - s - 2r - 2k. \quad (20)$$

An atom of real multiplicity $m \geq 2$ contributes $m - 2$ to E ; a nonreal pair of multiplicity m contributes $2(m - 1)$. Thus

$$O \leq s + E. \quad (21)$$

Indeed, a nonsimple odd real point has multiplicity at least three and contributes at least one to E . This is a statement about distinct odd points, not a bound for odd-multiplicity copies.

Proposition 2.3 (Exact parity and distinct residuals). *For the finite multiset above,*

$$3s \geq 2N - Q + 2O + D(G), \quad 6D_Z \geq 7N - 2Q + O + 2D(G). \quad (22)$$

Proof. The strong form already proved in Proposition 2.2 is

$$Q \geq 4N - 3s - 4r - 4k + D(G) = 2N - s + 2E + D(G).$$

Write its nonnegative slack as $\sigma = Q - (4N - 3s - 4r - 4k + D(G))$. Direct substitution gives the exact identities

$$3s - \{2N - Q + 2O + D(G)\} = 2(s + E - O) + \sigma, \quad (23)$$

$$6D_Z - \{7N - 2Q + O + 2D(G)\} = (s + E - O) + 6k + 2\sigma. \quad (24)$$

Both right-hand sides are nonnegative by (21). $\square$

For clarity, the atomic contributions to $s + E - O$ are zero at a simple real point, $m - 2$ at a real even point, $m - 3$ at a real odd point of multiplicity at least three, and $2(m - 1)$ at a nonreal pair. This proves the bookkeeping for all integer multiplicities without a finite cutoff. The parity inequalities are consequences of the known stable multiplicity bound, not a claim of a new rank–trace theorem.

Remark 2.4 (Sharp accounting and a false shortcut). The inequality $2D_Z \geq N + s$ is false for general multisets: one real point of multiplicity three has $(N, s, D_Z) = (3, 0, 1)$. The exact difference is $2D_Z - N - s = 2k - E$. Thus one cannot take one half of one plus the parity simple-zero bound to obtain (4).

Both residuals can vanish in an actual finite Hilbert model. Take the probability density $\mathbf{1}_{[-1/2, 1/2]}$ and distinct integer support points with multiplicities one or two. Their features are orthonormal, so $Q = s + 4r$, $N = s + 2r$, $O = s$, and $E = k = D(G) = \sigma = 0$. This is sharpness of the finite bookkeeping. It does not assert that zeta zeros realize these configurations.

One can also express the algebra as a scalar optimization. From normalized bounds $s - 2e \geq c$ and $s + e \geq o$, with $s, e, o \geq 0$, the exact minimum of s in this relaxation is

$$\max\{0, c, (c + 2o)/3\}. \quad (25)$$

The three terms are immediate lower bounds. Conversely, take s equal to the displayed maximum and $e = \max\{0, o - s\}$. If $s \geq o$, then $e = 0$ and $s \geq c$; otherwise $s - 2e = 3s - 2o \geq c$. This proves feasibility and the stated relaxation optimum. Extra total-mass or Gram constraints are not included, so no optimum over actual zero configurations is claimed.

3 From three-point energy to a density gain

For a positive semidefinite matrix M with diagonal one, define $\mathcal{E}(M) = 2 \sum_{i < j} |M_{ij}|^2$.

Lemma 3.1 (Block defect and pinching). *We have $D(M) \geq \min\{1, \mathcal{E}(M)\}$. If M_j are the principal blocks of a partition of M , then $D(M) \geq \sum_j D(M_j)$.*

Proof. If all eigenvalues of M are at most 2, then $D(M) = \text{tr}(M - I)^2 = \mathcal{E}(M)$. If some eigenvalue exceeds 2, its own Ψ value exceeds 1 and all other terms are nonnegative. For pinching, diagonalize each principal block and let (w_i) be the resulting orthonormal basis of the full space. Scalar Jensen applied to the spectral resolution of M gives

$$\sum_j D(M_j) = \sum_i \Psi(\langle w_i, M w_i \rangle) \leq \sum_i \langle w_i, \Psi(M) w_i \rangle = D(M).$$

Only scalar convexity of Ψ is used; no operator-convex assertion is needed. $\square$

Lemma 3.2 (Three shifted partitions). *Suppose $0 < d \leq 1$, $R > 0$, and the real even kernel satisfies*

$$2\{K(u)^2 + K(v)^2 + K(u+v)^2\} \geq d, \quad u, v \geq 0, \quad u+v \leq R. \quad (26)$$

If $\ell = x_s - x_1$ is the span of the simple real points, with $\ell = 0$ when $s \leq 1$, then

$$D(G) \geq \frac{d}{3} \left(s - 2 - \frac{2\ell}{R} \right). \quad (27)$$

Proof. For $s \geq 3$, the sum of the spans of its $s - 2$ consecutive triples is at most 2ℓ , since each successive gap occurs at most twice. Hence at most $2\ell/R$ triples have span greater than R . Every remaining triple has defect at least d by Lemma 3.1.

Separate the starting indices of consecutive triples into their three residue classes modulo 3. Within each class the triples are disjoint principal blocks; add the unused points as singleton blocks. Apply pinching to each partition and average the three inequalities. Every short triple has been counted once, proving (27). If its right-hand side is negative, nonnegativity proves the inequality; this also covers $s < 3$. $\square$

The factor $1/3$ records the three shifted partitions. Overlapping triples cannot simply be summed without this step.

3.1 Odd frames and exact pressure accounting

The next construction adds energies of pair-disjoint triples inside one frame. It applies defect pinching only between disjoint full frames. This distinction is essential: arbitrary overlapping triple defects cannot be summed. For instance, two consecutive three-by-three all-ones blocks have total defect ten, while their four-by-four all-ones parent has defect eight.

Lemma 3.3 (Odd-frame pressure bound). *Let K be a real even positive-type kernel with $K(0) = 1$, satisfying (10) with parameters $p, \varepsilon > 0$. Let $k \geq 1$, $M = 2k + 1$, and $k\varepsilon \leq 1$. For the Gram matrix of s ordered points of total span ℓ , set*

$$\alpha = \frac{k\varepsilon}{M}, \quad \beta = \frac{(M-1)p}{M} = \frac{2kp}{M}.$$

Then

$$D(G) \geq \alpha(s - M + 1) - \beta\ell. \quad (28)$$

As before, set $\ell = 0$ for $s \leq 1$.

Proof. For an M -point frame $y_1 \leq \dots \leq y_M$, select triples $(y_{2j-1}, y_{2j}, y_{2j+1})$, $1 \leq j \leq k$. They share at most one vertex and no unordered pair. Thus their doubled pair energies sum to at most $\mathcal{E}(G_F)$, while their spans telescope:

$$\sum_{j=1}^k (y_{2j+1} - y_{2j-1}) = y_M - y_1 =: \ell_F.$$

Adding their pressure inequalities gives $\mathcal{E}(G_F) + p\ell_F \geq k\varepsilon$. By Lemma 3.1,

$$D(G_F) + p\ell_F \geq k\varepsilon. \quad (29)$$

Indeed, if $\mathcal{E}(G_F) \leq 1$, then $D(G_F) \geq \mathcal{E}(G_F)$; otherwise $D(G_F) \geq 1 \geq k\varepsilon$. This is where the unit cap is used.

Partition the full sequence into complete frames with starts in one residue class modulo M , adding unused indices as singletons. Within each class the frames are disjoint. Across all M classes, there are exactly $(s - M + 1)_+$ complete frames, one for each possible start. Expanding their spans into adjacent gaps gives

$$\sum_{a=1}^{s-M+1} (x_{a+M-1} - x_a) \leq (M-1)\ell,$$

because each gap occurs at most $M-1$ times. Apply pinching and (29) in each class and average. This gives (28) for $s \geq M$. For $s < M$ its right side is nonpositive, so nonnegativity of D completes the proof. $\square$

The construction is a simple nonuniform pair-capacity schedule within the prior global pressure/frame methods of [8, 9, 11, 12]. Its short-interval consequence is obtained below from the exact finite operator, without an additional large-frame Gram approximation.

4 A quantitative obstruction for the cosine kernel

For $0 < \lambda \leq 1$ put

$$f_\lambda(u) = \frac{\cos(\sqrt{2}u)}{\sqrt{2}\sin(\lambda/\sqrt{2})} \mathbf{1}_{[-\lambda/2, \lambda/2]}(u), \quad K_\lambda = \widehat{f}_\lambda. \quad (30)$$

This is a nonnegative probability density. Set $a = \lambda/\sqrt{2}$, $q = a \tan a$, and $X = \pi\lambda t$. Integration yields

$$K_\lambda(t) = \frac{\cos X - (X/a) \cot(a) \sin X}{1 - X^2/a^2}. \quad (31)$$

The apparent singularities at $X = \pm a$ are removable. Equivalently, with $\text{sinc}(x) = \sin(x)/x$ continuously extended at zero,

$$K_\lambda(t) = \frac{\text{sinc}(X - a) + \text{sinc}(X + a)}{2 \text{sinc}(a)}. \quad (32)$$

An actual positive root satisfies $X \tan X = q$. If $X, Y, X+Y$ all corresponded to positive roots, tangent addition would force $X^2 + XY + Y^2 + q^2 = 0$. The following estimate gives a quantitative version without dividing by a possibly vanishing cosine or removable denominator.

Lemma 4.1 (Explicit positive triangle energy). *Fix $R > 0$, put $M = \pi\lambda R$, and define B, b as in (7), using $a = \lambda/\sqrt{2}$ and $q = a \tan a$. Then, throughout $u, v \geq 0$, $u + v \leq R$,*

$$2\{K_\lambda(u)^2 + K_\lambda(v)^2 + K_\lambda(u+v)^2\} \geq 2(b/B)^2 > 0. \quad (33)$$

Proof. Let $A_X = q \cos X - X \sin X$. For $Z = X + Y$, angle addition gives

$$(X^2 + XY + Y^2 + q^2) \sin X \sin Y = A_X A_Y - X A_X \sin Y - Y A_Y \sin X - q A_Z. \quad (34)$$

Take $X, Y \geq 0$, $Z \leq M$, and put $h = \max\{|A_X|, |A_Y|, |A_Z|\}$. If $h \leq q/2$, then

$$q \leq q|\cos X| + q|\sin X| \leq h + (q + X)|\sin X|,$$

so $|\sin X| \geq q/[2(q + M)]$, and likewise for Y . Taking absolute values in (34) therefore yields

$$\frac{q^4}{4(q + M)^2} \leq h^2 + (M + q)h \leq (M + 3q/2)h.$$

If $h > q/2$, the other alternative in the definition of b applies. In either case $h \geq b > 0$. The identity

$$A_X = q(1 - X^2/a^2)K_\lambda(X/(\pi\lambda))$$

holds also at the removable points by continuity. Since $|q(1 - X^2/a^2)| \leq B$ for $0 \leq X \leq M$, at least one of the three kernel values has absolute value at least b/B . This proves (33). $\square$

The zero-set obstruction itself is already used in [8]. The explicit formula in Lemma 4.1 makes its positive compact gap available for every fixed exponent without a numerical minimum. Its small size is acceptable for the qualitative strict-improvement theorem; Section 7 certifies a more useful gap in one example.

5 Short-interval arithmetic and passage to the sharp density

We record the exact analytic interface to avoid a support or smoothing uniformity assumption. Fix $0 < \lambda < \theta < 1$, put $H = T^\theta$, $L = \log T$, and $I = (T, T + H]$. Wang's Theorem 2.2 states that for every fixed real even $g \in C_c^\infty(\mathbb{R})$ supported in $[-\lambda, \lambda]$,

$$\begin{aligned} W_I(g) &:= \sum_{\gamma, \gamma' \in I} \hat{g}\left(\frac{i(\rho - \rho')L}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} \\ &= \frac{HL}{2\pi} \left(g(0) + \int_{\mathbb{R}} |\alpha| g(\alpha) d\alpha \right) + O_g(H + T^\lambda L^2). \end{aligned} \quad (35)$$

The constants may depend on fixed λ, θ . This is an unconditional complex-zero formula; RH is not an assumption.

Choose a fixed real even $\eta \in C_c^\infty((-\lambda/2, \lambda/2))$ with $\int \eta^2 = 1$, and write $f = \eta^2$, $K = \hat{f}$, $g = f * f$. Both g and g'' meet the fixed-test hypotheses. At $z = i(\rho - \rho')L/(2\pi)$, differentiation of the Fourier transform gives

$$\widehat{g''}(z) = -4\pi^2 z^2 K(z)^2 = L^2(\rho - \rho')^2 K(z)^2.$$

The exact identity

$$\left(\widehat{g}(z) - \frac{\widehat{g''}(z)}{4L^2} \right) \frac{4}{4 - (\rho - \rho')^2} = K(z)^2$$

removes the rational weight. Apply (35) separately to the two fixed tests and combine the resulting estimates. As in [3, Lemma 3.1], this gives

$$\sum_{\gamma, \gamma' \in I} K(z_\rho - z_{\rho'})^2 = \mathcal{C}(f) \frac{HL}{2\pi} + O_f(H + T^\lambda L^2), \quad (36)$$

where

$$z_\rho = \frac{i(\rho - 1/2)L}{2\pi}, \quad \mathcal{C}(f) = \int f^2 + \iint |u - v| f(u) f(v) du dv. \quad (37)$$

Indeed, the main term for g'' is $-\int (f')^2 + 2\int f^2$, finite for the fixed smooth density, and its coefficient is $1/(4L^2)$. A varying scalar coefficient after applying the theorem to two fixed tests does not require a varying-test theorem.

Reflection $\rho \mapsto 1 - \bar{\rho}$ preserves the height interval and multiplicity and maps z_ρ to $\bar{z}_\rho$. Thus the multiset $Z_T = \{z_\rho : \gamma \in I\}$ satisfies Proposition 2.2. Its simple real points are precisely the simple critical zeros. Their real-coordinate span is at most $X_T = HL/(2\pi)$. The short-interval Riemann–von Mangoldt formula gives

$$N(T, H) = X_T + O(H + \log T), \quad X_T/N(T, H) \rightarrow 1. \quad (38)$$

After division by N , the error in (36) is $O_f(1/L + T^{\lambda-\theta}L) = o(1)$. The strict support gap $\lambda < \theta$ is therefore necessary at this stage.

The cosine density (30) is Wang’s minimizing density and satisfies

$$\mathcal{C}(f_\lambda) = \frac{\lambda}{2} + \frac{1}{\sqrt{2}} \cot(\lambda/\sqrt{2}) = 2 - c(\lambda). \quad (39)$$

It is not a smooth compactly supported test because its endpoint values do not vanish. Choose even cutoffs $0 \leq \chi_\varepsilon \leq 1$, supported strictly inside $(-\lambda/2, \lambda/2)$ and approaching one in the interior. Normalizing $\eta_\varepsilon = \chi_\varepsilon \sqrt{f_\lambda} / \|\chi_\varepsilon \sqrt{f_\lambda}\|_2$ gives densities $f_\varepsilon = \eta_\varepsilon^2$ converging in both L^1 and L^2 to f_λ . Moreover $f_\lambda \rightarrow f_\theta$ in those spaces as $\lambda \uparrow \theta$, and $\mathcal{C}$ is continuous for these convergences on bounded support [3, Section 4].

For normalized nonnegative densities,

$$\sup_{x \in \mathbb{R}} |\hat{f}(x) - K_\theta(x)| \leq \|f - f_\theta\|_1, \quad |\hat{f}(x)|, |K_\theta(x)| \leq 1. \quad (40)$$

Thus the doubled triple energy changes by at most $12\|f - f_\theta\|_1$. This retains a fixed positive energy margin through the smooth approximation.

Proposition 5.1 (Transfer with a certified limiting gap). *Fix $0 < \theta < 1$, $R > 0$, and $0 < d \leq 1$. Suppose (26) holds for K_θ . Define*

$$c_d = \frac{c(\theta) - 2d/(3R)}{1 - d/3}.$$

Then the simple-critical and distinct proportions in (9) are at least c_d and $(1 + c_d)/2$, respectively.

Proof. First fix $0 < d' < d$. Choose a sequence of fixed smooth densities f_j supported in $(-\lambda_j/2, \lambda_j/2)$, with $\lambda_j < \theta$, converging to f_θ in L^1 and L^2 . By (40), every sufficiently large j has triple energy at least d' . Fix such a j before letting T grow.

Write $s = N_0^s(T, H)$, $N = N(T, H)$, and Q for the pair sum with this fixed kernel. Proposition 2.2 and Lemma 3.2, with actual span at most X_T , imply

$$\left(1 - \frac{d'}{3}\right) \frac{s}{N} \geq 2 - \frac{Q}{N} - \frac{2d'}{3N} - \frac{2d'X_T}{3RN}.$$

Let $T \rightarrow \infty$, using (36) and (38), and then let $j \rightarrow \infty$. The result is $\liminf s/N \geq c_{d'}$.

For the distinct count, the other finite inequality gives

$$\liminf \frac{N^d(T, H)}{N} \geq \frac{1 + c(\theta)}{2} + \frac{d'}{6}(c_{d'} - 2/R) = \frac{1 + c_{d'}}{2}.$$

The equality uses $c_{d'} - c(\theta) = d'(c_{d'} - 2/R)/3$. Finally let $d' \uparrow d$; all scalar expressions are continuous since $d \leq 1 < 3$. No test function changes inside a large- T limit. $\square$

Proof of Theorem 1.2. Lemma 4.1 with $\lambda = \theta$ proves a limiting-kernel gap at least $2(b/B)^2$, and hence at least the positive δ in (7). Apply Proposition 5.1 with $d = \delta$. Its constant equals (8). The gain is strictly positive because $R > 2/c(\theta)$. $\square$

Proof of Theorem 1.3. First fix $0 < \varepsilon' < \varepsilon$, keeping p and k fixed. Choose fixed smooth densities f_j supported inside $(-\lambda_j/2, \lambda_j/2)$, with $\lambda_j < \theta$, converging to f_θ in L^1 and L^2 . The uniform estimate (40) changes the triple energy by at most $12\|f_j - f_\theta\|_1$. Thus every sufficiently large j satisfies the all-gap pressure inequality with the same p and target ε' . Also $k\varepsilon' < 1$.

Put $M = 2k + 1$, $\alpha' = k\varepsilon'/M$ and $\beta = 2kp/M$. For each individual fixed j , Proposition 2.2 and Lemma 3.3 give

$$(1 - \alpha')\frac{s}{N} \geq 2 - \frac{Q}{N} - \beta\frac{X_T}{N} - \frac{\alpha'(M - 1)}{N}.$$

Take $T \rightarrow \infty$ with this test fixed. If $c_j = 2 - \mathcal{C}(f_j)$ and $b_j = (c_j - \beta)/(1 - \alpha')$, then $\liminf s/N \geq b_j$. The stronger finite inequality for distinct points gives

$$2 \liminf \frac{N^d(T, T^\theta)}{N} \geq 1 + c_j + \alpha'b_j - \beta = 1 + b_j.$$

The identity follows from $(1 - \alpha')b_j = c_j - \beta$. It does not use the false general counting shortcut $N^d \geq (N + s)/2$.

Now let $j \rightarrow \infty$, so $c_j \rightarrow c(\theta)$, and finally let $\varepsilon' \uparrow \varepsilon$. The limiting denominator is at least $M - 1 > 0$, including the endpoint $k\varepsilon = 1$. This proves both proportions in Theorem 1.3. No support parameter or smoothing scale varies within a height limit. $\square$

Proof of Corollary 1.4. Lemma 4.1 gives the compact energy bound with $d = \delta$. Nonnegativity outside its triangle gives the all-gap inequality

$$2\{K_\theta(u)^2 + K_\theta(v)^2 + K_\theta(u + v)^2\} + (d/R)(u + v) \geq d.$$

Apply Theorem 1.3 with $\varepsilon = d$ and $p = d/R$. The gain coefficient satisfies

$$\frac{kd}{2k + 1 - kd} - \frac{d}{3 - d} = \frac{d(k - 1)}{(2k + 1 - kd)(3 - d)} > 0$$

for $k > 1$. Finally $b \leq q/2$ and $B > q$ in (7), so $2(b/B)^2 < 1/2$ and $0 < \delta = (b/B)^2 < 1/4$. Thus $k = 2$ is always admissible, giving the stronger entire positive curve. $\square$

The same compact-to-pressure conversion works with any stronger certified d , using strict lower targets during smoothing before the final scalar limit. In particular $d = 1/7000$, $R = 21/4$, $k = 7000$ gives (12). Its gain relative to the original triple bound is exactly $(3 - d)/2 = 20999/14000$. This is a ratio of gains above Wang's baseline, not a ratio of the full proportions.

6 Classical odd zeros and the range extension

The positive Gram defect is useful for quantitative pressure improvements. For the qualitative parity theorem we may discard it: the multiplicity excess in the same finite inequality already supplies the missing information when combined with a classical sign-change theorem.

6.1 The exact classical count and interval packing

We use the following fixed consequence of Selberg's theorem, stated as Theorem B on p. 523 of Karatsuba [4]: for the fixed $\alpha = 51/100$ there exist constants $a > 0$ and t_a such that

$$N_{\text{odd}}(t, t^\alpha) \geq at^\alpha \log t \quad (t \geq t_a). \quad (41)$$

The zeros in this count are distinct and of odd order on the critical line. Karatsuba's stronger main theorem, p. 524, gives the same kind of bound for $\alpha = 27/82 + \epsilon$, $0 < \epsilon \leq 0.001$. His final argument in Section 12, p. 536, selects disjoint sign-changing intervals; this confirms the distinct-count convention. We import the classical theorem and its analytic estimates; we do not reprove them. Selberg's original paper was not consulted, so our source for its exact statement is Karatsuba's primary article.

Fix $\alpha < \theta < 1$ and put $H = T^\theta$. Starting at $t_0 = T$, define $t_{j+1} = t_j + t_j^\alpha$ and keep all complete adjacent intervals lying inside $(T, T + H]$. Their total length is $H - O(T^\alpha) = H(1 - o(1))$. Their endpoints are all $T(1 + o(1))$, and hence their logarithms are $\log T + o(1)$. Summing (41) gives

$$N_{\text{odd}}(T, H) \geq (a + o(1))H \log T. \quad (42)$$

If a seed is stated with different open or closed endpoint conventions, at most two distinct points per block must be removed. There are $O(H/T^\alpha + 1)$ blocks, so this loss is $o(H \log T)$ and does not change the conclusion. Counting odd copies instead would not justify this bounded endpoint loss, which is another reason to keep the precise classical convention.

The Riemann–von Mangoldt formula gives $N(T, H) \sim H \log T / (2\pi)$. Choose once and for all a sufficiently small $\kappa > 0$ with $\kappa < 2\pi a$. Then

$$\liminf_{T \rightarrow \infty} \frac{N_{\text{odd}}(T, T^\theta)}{N(T, T^\theta)} \geq \kappa \quad \text{for every fixed } \alpha < \theta < 1. \quad (43)$$

The same κ works because it was chosen from the one fixed seed. Only the eventual height is allowed to depend on θ . Nothing here permits $\theta - \alpha$ to shrink with T .

6.2 Fixed-test transfer, including nonpositive cosine bounds

Fix $\alpha < \theta < 1$ and $0 < \lambda < \theta$. Use the transformed zeta multiset and one fixed smooth probability density $f = \eta^2$ with support in $(-\lambda/2, \lambda/2)$ from the arithmetic transfer above. Its real simple and odd points are precisely those counted by N_0^s and N_{odd} . Equations (36) and (37) give

$$Q/N \longrightarrow \mathcal{C}(f).$$

This is an exact unweighted pair sum followed by an asymptotic limit. The weight removal uses Wang's theorem separately for two fixed test functions, including the differentiated test; it does not assume that individual complex pair terms are nonnegative. Its normalized error is $O_f(1/\log T + T^{\lambda-\theta} \log T) = o(1)$.

Apply Proposition 2.3, Proposition 2.2, $D(G) \geq 0$, $N^d \geq N_{\text{odd}}$, and (43). For this fixed test we obtain

$$\begin{aligned} \liminf N_0^s/N &\geq \max\{0, 2 - \mathcal{C}(f), [2 - \mathcal{C}(f) + 2\kappa]/3\}, \\ \liminf N^d/N &\geq \max\{\kappa, [3 - \mathcal{C}(f)]/2, [7 - 2\mathcal{C}(f) + \kappa]/6\}. \end{aligned}$$

Choose the normalized squares of smooth cutoffs of the cosine square root described earlier, with fixed support parameter λ . After the height limit, let the cutoff converge in L^1 and L^2 to f_λ . Then $\mathcal{C}(f) \rightarrow 2 - c(\lambda)$ by (39). Finally let $\lambda \uparrow \theta$. Continuity gives (3)–(4). This argument is valid when $c(\lambda) \leq 0$; positivity was needed for the pressure gain, not for the pair-sum limit or the finite inequalities.

For completeness, the cosine cost also has a short direct evaluation. Let $J(x) = \int |x - y| f_\lambda(y) dy$. On the support interior, $J'' = 2f_\lambda$ and $f_\lambda'' = -2f_\lambda$. The even function $f_\lambda + J$ is therefore constant. At $b = \lambda/2$ it equals $b + \cot(\lambda/\sqrt{2})/\sqrt{2}$, since $J(b) = b$. Multiplication by f_λ and integration gives

$$\mathcal{C}(f_\lambda) = \frac{\lambda}{2} + \frac{1}{\sqrt{2}} \cot(\lambda/\sqrt{2}) = 2 - c(\lambda).$$

The legal order is thus fixed test, height limit, smooth approximation, and support limit. No test or derivative degree grows with the height.

6.3 Proof of the strict range statements

The definition of θ_0 and the derivative formula require no computed root. To verify $\alpha < \theta_0$, put $a_0 = \alpha/\sqrt{2}$. The elementary bounds $\cos a_0 \geq 1 - a_0^2/2 > 0$ and $0 < \sin a_0/a_0 \leq 1$ imply

$$a_0 \cot a_0 \geq 1 - \alpha^2/4, \quad c(\alpha) \leq 2 - \frac{1}{\alpha} - \frac{\alpha}{4} = -\frac{1801}{20400} < 0.$$

Also $\tan x > x$ on $(0, \pi/2)$ gives $c(1) > 1/2$, and $c'(\theta) > 0$ gives the unique zero. On $[\alpha, \theta_0]$,

$$c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) < \frac{1}{\theta^2} \leq \frac{10000}{2601} < 4. \quad (44)$$

For δ in (5), we have $\alpha < \theta_1 < \theta_0$ and

$$c(\theta_1) = - \int_{\theta_1}^{\theta_0} c'(t) dt > -4\delta \geq -\kappa.$$

The right-hand side of (3) is therefore greater than $\kappa/3$ at θ_1 , and is no smaller at any fixed larger exponent. With δ_{half} in (6), the same argument gives $c(\theta_{\text{half}}) > -\kappa/4$; the separately proved distinct formula yields

$$\frac{3 + 2c(\theta_{\text{half}}) + \kappa}{6} > \frac{1}{2} + \frac{\kappa}{12}.$$

This proves the remaining claims of Theorem 1.1.

At θ_0 itself, (3) gives a lower proportion at least $2\kappa/3 > 0$. The strict extension follows from this positive margin and continuity, without a new pair-correlation support estimate. Its numerical size remains unknown because the imported fixed classical constant has not been evaluated.

7 The compact certificate

For $\theta = 3/4$ and $R = 21/4$, a finite interval computation proves

$$2\{K_{3/4}(u)^2 + K_{3/4}(v)^2 + K_{3/4}(u+v)^2\} \geq \frac{1}{7000} \quad (u, v \geq 0, u+v \leq 21/4). \quad (45)$$

The certificate is a complete binary partition of the square $[0, R]^2$. Every endpoint and threshold is rational. A box is either subdivided, proved wholly outside the triangle, or accepted using an

energy lower bound valid on the entire box. The final tree contains 48,761 nodes, 24,252 accepted energy leaves, 129 outside leaves, and no unresolved boxes (computation EXP-002 [14]).

For completeness, the analytic enclosure used at each leaf is simple. Since f_θ is a probability density on $[-\theta/2, \theta/2]$,

$$|K'_\theta(x)| \leq 2\pi \int |u|f_\theta(u)du \leq \pi\theta. \quad (46)$$

On a real interval with center x_0 and radius r , an outward interval evaluation of $K_\theta(x_0)$ therefore supplies the valid lower bound

$$|K_\theta(x)| \geq \max\{0, \text{lower}(|K_\theta(x_0)|) - \pi\theta r\}.$$

Apply this separately to the box ranges for u , v , and $u+v$, square the nonnegative lower bounds, and double their sum. Every acceptance comparison requires the entire lower enclosure to exceed the rational threshold. An outside box is discarded only if the sum of its two lower coordinates is strictly greater than R , so the triangle boundary is retained.

Discovery used Arb through `python-flint==0.9.0` at 160-bit precision and the removable-singularity-free formula (32). Each accepted leaf was replayed at 256 bits with a separately derived Taylor evaluator. For real x ,

$$\text{sinc}(x) = \sum_{j=0}^{95} \frac{(-1)^j x^{2j}}{(2j+1)!} + \mathcal{R}(x), \quad |\mathcal{R}(x)| \leq \frac{|x|^{192}}{193!}. \quad (47)$$

The remainder follows by applying the real sine Taylor theorem through degree 192 and dividing by x , with continuous extension at zero. Arb encloses both the polynomial and its remainder. This replay is an independent kernel evaluator; it shares the partition reconstruction, Lipschitz argument, and Arb library with the first computation. It is not a second independent full verifier or a Lean proof.

Proof of Corollary 1.5. Use the finite certificate (45) in Proposition 5.1, with $d = 1/7000$ and $R = 21/4$. The full threshold d is available because the proposition first uses every $d' < d$ through smoothing and then passes to d . Substitution gives the stated constants. The outward rational enclosures for their values and positive difference are recorded with the certificate. $\square$

Proportion at $\theta = 3/4$	Wang baseline	This certificate
Simple and critical	0.4190750129754243...	0.4190768284253039...
Distinct	0.7095375064877121...	0.7095384142126519...

Table 1: All denominators count nontrivial zeros with multiplicity in $(T, T + T^{3/4}]$. The entries are proportions, not percentages.

The certificate’s role is quantitative. Even without it, Theorem 1.2 proves a strict improvement for every fixed $\theta > \theta_0$ using its explicit analytic gap. Conversely, the finite numerical inequality alone does not prove a theorem about zeta zeros without the finite operator and fixed-test analytic transfer.

8 Conclusions and further questions

Finite stability retains two different kinds of information. The Gram defect yields quantitative gains when simple critical points occur close enough for certified pressure windows to interact. Multiplicity excess, combined with distinct odd-multiplicity density, yields a qualitative range extension even at the zero of the baseline cosine bound. The two mechanisms have separate hypotheses, certificates, and distinct-zero consequences; neither proves the Riemann hypothesis.

The next analytic task for the parity route is to make one classical seed constant explicit enough to evaluate θ_1 . A further localized short-mollifier argument might supply stronger seed information, but the necessary general derivative theorem and precise zero count must first be proved. For the pressure route, richer windows may improve constants within the established framework; their existence does not establish global optimality. These are open tasks rather than conclusions of this paper.

Declaration of generative AI use. Generative AI tools were used for derivation, source research, review, and implementation. The author reviewed and verified all content and takes full responsibility for the work.

A Reproducibility, provenance, and scope

The repository github.com/fsantibanezleal/CAOS_RESEARCH archives each computation cited here under an identifier EXP-nnn, in the folder whose name starts with that identifier under `problems/number-theory/riemann-hypothesis/experiments/`. Each such folder contains the statement tested, the proof, the code, its outputs, and the verification of the result. The folder of computation EXP-002 [14] contains the mathematical proof, an independent verification of the proof, the interval runner, the canonical partition tree, and rational enclosures for the constants. A deterministic floating grid selected candidate radii; its sampled values were never used as certified lower bounds. The final certification used CPU arithmetic; no GPU was necessary.

From the repository root, with the problem’s pinned requirements installed, a fresh output directory preserves the canonical artifacts:

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-002-short-interval-stability/run.py \
    --output-dir tmp/riemann-replay --radius 21/4 --threshold 1/7000
pytest tests/test_riemann_certificates.py
```

The displayed backslashes indicate line continuation for readability; use one command line on shells with different continuation syntax. The runner fails with an exact resumable checkpoint if its time or node limit is reached while boxes remain. It does not report a certificate in that state.

The tree replay checks its hash, reconstructs every box, verifies all accepted leaves, and rejects missing, trailing, unknown, or miscounted nodes. A hash protects accidental corruption; geometric replay is still required when a modified tree has a recomputed hash. Negative controls confirmed the rejection of false whole-square acceptance, a false outside-domain claim, truncation, appended data, altered counts, an unresolved flag, an invalid exponent, and an inflated threshold. The focused tests include one complete 256-bit certificate replay. These are implementation checks, not formal verification of the entire analytic proof.

The canonical preorder tree has SHA-256 digest

78829635026aeacd69c02213cc899b5029ef2566bbfaf1c86b757cc5f92ffe55

B The two-variable pressure certificate

For the rational parameters (13), pressure alone proves (10) when $u + v \geq R_{\text{cut}} = \varepsilon/p = 443239/80000$. A finite interval computation certifies the remaining closed triangle. It reconstructs a complete binary partition of the square $[0, R_{\text{cut}}]^2$. Each leaf is either accepted by an outward-rounded energy-plus-pressure bound on its entire rectangle, or by the exact inequality $p(u_0 + v_0) \geq \varepsilon$ at its lower corner. Equality at the cutoff is permitted. Outside the square, pressure alone suffices.

The energy enclosure is the one justified in (46); the pressure contribution on a box is bounded below by $p(u_0 + v_0)$. The certificate has 16,797 nodes, 8,351 validated energy-plus-pressure leaves, 48 pressure-only leaves, and zero unresolved boxes. Construction uses Arb at 160 bits. Every accepted leaf is replayed at 256 bits with the separately derived sinc-Taylor evaluator. These evaluators share Arb, geometric subdivision, and the Lipschitz estimate; this is not a wholly independent verifier or a formal proof of the analytic theorem.

Both bounds of Corollary 1.6 are established in computation EXP-003 [15]. For C_A in (12), the computation replays the 48,761-node compact certificate of Section 7 completely at 256 bits, in about 26.9 seconds, and checks two symbolic identities and 328 small-index incidence/boundary cases. For C_B , it constructs the two-variable certificate described above and replays every accepted leaf, in about 5.2 seconds in total. The gain of C_B above Wang’s bound is

$$C_B - c(3/4) = 0.000012875194687394224537573076 \dots,$$

and it exceeds 5/4 of the gain of C_A :

$$(C_B - c(3/4)) - \frac{5}{4}(C_A - c(3/4)) = 0.000009471388256765351202430110 \dots > 0.$$

Together with $k\varepsilon = 999947184/10^9 < 1$, the finite certificate and Theorem 1.3 prove the statements on C_B in Corollary 1.6. Floating-point sampling was used only to choose the rational parameters (13), which are not claimed to be optimal; every inequality in the certificates is certified in exact rational or ball arithmetic.

Construction checkpoints are candidate proof state; a completed certificate is accepted only after the final arithmetic replay, and a replay restarted from a checkpoint checks the checkpoint identity and revalidates its entire prefix. The focused tests include tampering controls: a forged completed prefix with recomputed hashes is rejected by the leaf arithmetic.

The canonical result, the pressure proof tree, provenance hashes, and the exact rational endpoints are archived in the EXP-003 record [15]. Mathematical artifacts use deterministic UTF-8/LF bytes; elapsed-time logs are separate metadata. To reproduce with the repository virtual environment, run each command with a fresh output directory. The first command recomputes C_A , the second tests checkpointing and restart, and the third runs the focused tests:

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-003-odd-frame-pressure/run.py --stage=A \
    --output-dir tmp/riemann-a
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-003-odd-frame-pressure/run.py --stage=smoke \
    --output-dir tmp/riemann-smoke
python -m pytest tests/test_riemann_pressure.py
```

The full command line for the two-variable certificate is in the EXP-003 record [15]. The displayed backslashes are typographical line continuations; combine each command on shells that use another continuation convention.

C Parity verification and its analytic boundary

The EXP-004 record [16] separates the exact certificates (23)–(24), universal atom formulas, finite regression censuses, the scalar relaxation (25), sharpness examples, and the check of the analytic sources. No interval optimization, zeta-zero sampling, or GPU computation is needed for this route. In particular, the classical κ and the new numerical exponent are deliberately unevaluated fields, not outputs of a finite census.

Independent checks of the proof of the qualitative range theorem, within the stated analytic inputs, cover the signed finite operator, multiplicity charges, distinct-odd convention, interval packing, exact deweighting, and legal limiting sequence. A separate [proof-review file](#) lists the SHA-256 hashes of the proof, the proof checks, the statement tested, the computed result, and the summary of the outcome.

The arithmetic run took about 8.04 seconds. It verified two symbolic residual identities and nonnegative arbitrary-multiplicity atom families, supplemented by 24 single-atom cases through multiplicity 12. The full census consists of 19,683 count vectors in $\{0, 1, 2\}^9$, with 59,049 evaluations at three nonnegative slack values. All 42 rational scalar-relaxation cases have matching feasible primal and dual witnesses. All 36 finite sinc simple/double-support examples attain equality, while a real triple point rejects the false distinct half-sum with residual -1 . The exact derivative identity and rational comparisons in (44) passed.

The focused software tests check tampered counts, the exact bytes of the committed source files, preservation of an incomplete census prefix, input rejection, and protection of existing outputs against overwriting. In two independent recomputations, every stored census row was parsed with direct formulas without importing the runner; all count vectors, residuals, linked artifact hashes, and intentionally null numerical constants matched. The original run outputs are archived unchanged in the experiment record. Operational timing is kept separate from deterministic mathematical bytes.

The canonical result has SHA-256 digest

bc9e28ca1792657e24da77268e6e2108c3f52684b7e16504f07b32cfce014d6e

To reproduce the finite computation in a fresh directory, use

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-004-parity-density-transfer/run.py \
    --output tmp/riemann-parity-replay
pytest tests/test_riemann_parity.py
```

The displayed backslashes are typographical line continuations. The run verifies the algebra and the finite controls above in exact arithmetic. The universal atom argument, Selberg’s theorem, interval packing, Wang’s fixed-test transfer, and the approximation limits supply the unbounded quantifiers. A finite computation cannot establish those analytic statements by enumeration. The census slack samples are not claimed to be geometric realizations of zeta zeros, and the scalar optimum concerns only the stated relaxation.

D Source comparison and limitations

The source comparison was completed on 2026-09-12. It used the full Wang v1 manuscript, Lamzouri v2, the current Alpöge–Furman paper, and pinned snapshots of the directly relevant global

stability, pressure-frame, Bellman-coboundary, and mixed-certificate repositories. Queries combining the Wang identifier, short intervals, pressure, stability, Gram defects, and exact numerical signatures found no prior public statement of these short-interval refinements. The exact odd-frame schedule was not located, but is a simple instance of known global pair capacities and pressure methods. The globally asserted spectrum-energy envelope is excluded from new-theorem claims. This supports the scoped contribution claimed here, without establishing exhaustive priority or detecting unpublished concurrent work. The known finite stability inequality and cosine-root mechanism are explicitly credited to the repository identifier `ainta`; no personal author identity is inferred from that identifier.

For the parity theorem, the source comparison also covered Karatsuba’s complete primary paper, Steuding’s archived dissertation and published abstract, and *Short mollifiers of the Riemann zeta-function* [7]. The short-mollifier result concerns critical-zero density using higher derivatives; it supplies no automatic simple-zero theorem. Steuding’s stated localized theorem uses the first derivative. His proof suggests a route to fixed higher derivatives, but a uniform two-shift extension, the detector’s exact counting convention, and a certified explicit seed would all require additional proof. None is assumed here. Likewise, weighted all-critical Gram defects and negative off-line spectral mass are separate research directions, not premises of the parity theorem.

A bounded search of primary literature and the pinned contributor sources did not locate the exact short-interval parity combination. The finite accounting is already a consequence of known general multiplicity inequalities; the claimed contribution is its stated and proved asymptotic consequence with a classical odd-zero seed. This is not an assertion of exhaustive priority or a replacement for specialist peer review. In particular, no numerical improvement on the best explicit published positivity exponent is asserted.

The analytic inputs include Wang’s recent preprint and the classical sign-change theorems, not an end-to-end machine formalization. The mathematical and numerical reviews can still contain errors, and the two numerical evaluators share arithmetic and geometric components. The parity theorem gives an existentially smaller fixed exponent; it does not give its decimal value, an effective starting height, uniformity for moving exponents, a global-record proportion, or elimination of all off-line zeros.

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